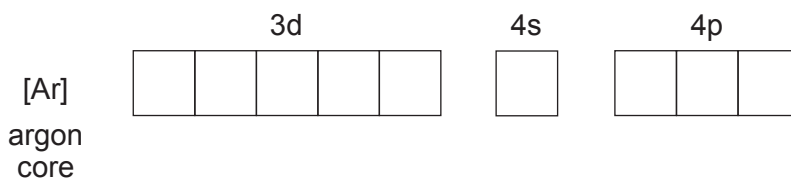


SECTION A

Answer **all** questions in the spaces provided.

1. Using the convention of arrows to represent electrons, complete the electronic structure of an arsenic atom, As. [1]



2. Give the oxidation number of molybdenum in the ion MoO_4^{2-} . [1]

.....

3. The half-life of the radioactive isotope oxygen-15 is 2 minutes.

- (a) Give the atomic number, mass number and symbol of the nuclide produced when 1 atom of ^{15}O decays by the emission of one positron. [1]

.....

- (b) If a quantity of ^{15}O decays by positron emission, determine what fraction of ^{15}O is left after 10 minutes. [1]

.....

4. Explain why the second molar ionisation energy of sodium is greater than its first molar ionisation energy. [1]

.....
.....
.....



SECTION A

Answer all questions in the spaces provided.

1. Using **outer** electrons only, draw a dot and cross diagram to show the bonding in calcium bromide. [2]

2. Identify the **two** elements from the following list that together produce the most ionic bond. Explain your choice.

bromine

magnesium

oxygen

sodium

[2]

.....

.....

.....

3. (a) State how a coordinate bond differs from a covalent bond. [1]

.....

.....

- (b) Give **one** example of a species containing a coordinate bond. [1]

.....



SECTION A

Answer all questions in the spaces provided.

1. For the ionic compound caesium chloride, state the coordination number of the chloride ion. [1]

.....

2. The first four ionisation energies for an element are shown in the table below.

Ionisation energy / kJ mol^{-1}			
1st	2nd	3rd	4th
548	1060	4120	5440

State to which group in the Periodic Table the element belongs.

[1]

.....

3. By inserting arrows to represent electrons, complete the electronic structure of a chromium atom. [1]

	3s	3p	3d	4s
neon core				



SECTION A

Answer **all** questions in the spaces provided.

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in magnesium fluoride. [2]

2. In some areas, fluoride ions are added to drinking water.

State **one** benefit of adding fluoride ions to water. [1]

.....

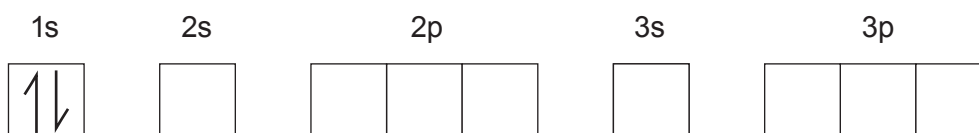
.....

3. Complete the following definition of relative atomic mass. [1]

The relative atomic mass of an element is the average mass of one atom of the element relative to

.....

4. Sodium forms only one stable ion. By inserting arrows to represent electrons, complete the electronic structure of this ion. [1]



SECTION A

Answer **all** questions.

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in sodium oxide. [2]

2. Complete the following sentence: [1]

Hydrogen bonding is an intermolecular force that occurs between molecules containing hydrogen atoms bonded to small, atoms which have lone pairs of electrons for example

3. By inserting arrows to represent electrons, show the electronic configuration of a calcium atom. [1]

1s	2s	2p	3s	3p	3d	4s

4. Hydrazine can be manufactured from ammonia.



Calculate the atom economy of this reaction. [1]

atom economy = %



SECTION AAnswer **all** questions.

1. Complete the table below.

[2]

Molecule	Number of bonding pairs of electrons in outer shell	Number of lone pairs of electrons in outer shell	Shape
BeCl ₂	2	0	
PCl ₃			pyramidal

2. Complete the table below to show the type or types of bonding present in the following solids.

[2]

Solid	Type or types of bonding
calcium	
iodine	

3. Give the oxidation number of rhenium in ReOCl
- ₄
- .

[1]

.....

